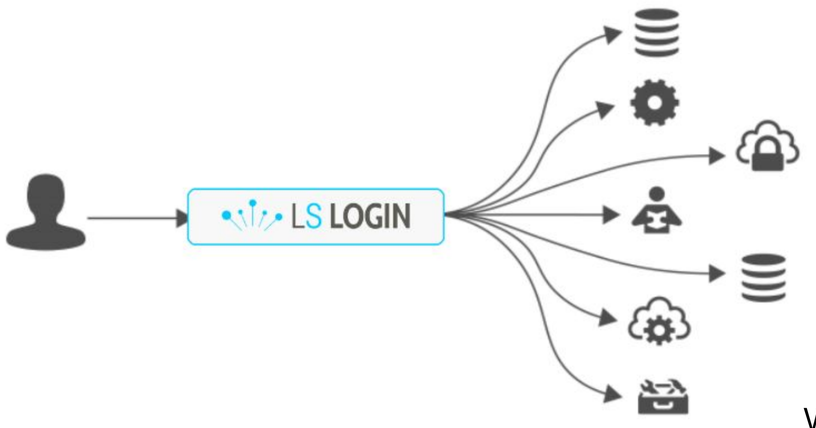
Modernising Life Sciences: ELIXIR Compute Platform's Vision for 2024-2026

ELIXIR All Hands , 10-12 June 2024, Uppsala, Sweden

Life Science Login

Advanced service access control

This work will develop advanced authentication and authorisation mechanisms, aligning with GA4GH standards such as Passports for compliant sensitive data access. The work includes enhancing authorisation mechanisms across distributed infrastructures and improving user experience. The LS Login authentication process will adopt more secure mechanisms, while authorisation efforts will concentrate on efficiently defining and applying rules. These enhancements will bolster LS Login usage in key projects like GDI.



Trusted Research Environments

Protected access to sensitive data for analysis

Sensitive data plays a critical role in research and can accelerate scientific discovery. This work will extend and align ELIXIR efforts in this space such as breaking down sensitive data access into a process to comprehend the general workflow across ELIXIR Nodes through use cases. Reference implementations of protected access to sensitive data, incorporating common steps and services will also be developed. Guidance and outreach efforts will ultimately be developed to assist ELIXIR members seeking support on protected data access for analysis in alignment with the EOSC ENTRUST project.

Sensitive data access workflow | **Guidance & outreach** | **TRE alignments**

eosc | ENTRUST
European Network of Trusted Research Environments

ELIXIR Cloud

Will integrate Compute efforts to leverage a coordinated network of ELIXIR Nodes to deliver a GA4GH standards-compliant federated analytics environment.

Learn more about ELIXIR Cloud:

Platform Leadership

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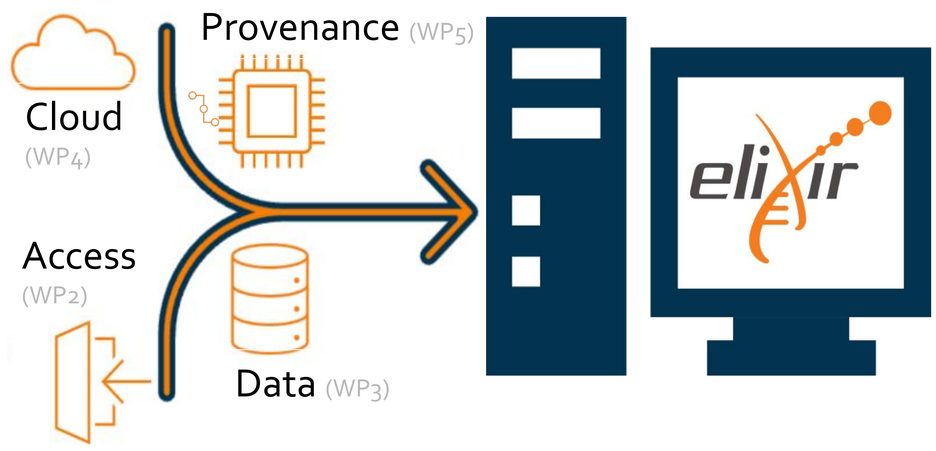
Platform Mission

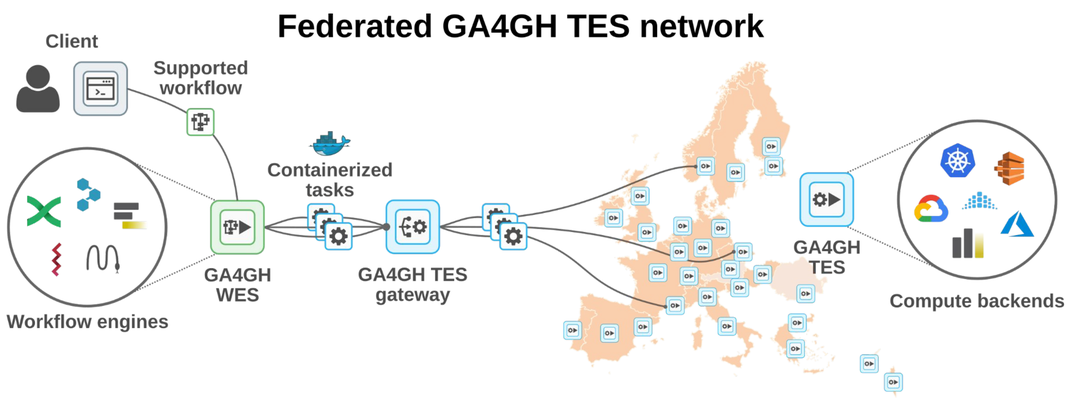
To coordinate and integrate cloud, compute, storage and access services for the life-science research community.



Interoperable Cloud

Multi cloud infrastructure deployment

This work aims to advance an ELIXIR-wide federated hybrid- and multicloud-ready package composed of various GA4GH-compatible backend microservices deployed across ELIXIR Nodes. Key activities include expanding the ELIXIR Cloud Software Development Kit (SDK) use case portfolio and initiating the operationalisation of the ELIXIR Cloud by deploying centralised and federated ELIXIR Cloud SDK service components across different ELIXIR Nodes.



Accounting & Provenance

Sustainability, accounting and provenance for federated analytics

This work aims to enhance life science analytics by developing a resource management framework that prioritise sustainability, accountability, and provenance. Key activities include creating a live dashboard for ELIXIR Computing Providers to aid in compute resource findability, and utilising RO-Crate formats to aid research reproducibility and usage accounting. The project focuses on identifying best practices in reporting resource usage across ELIXIR Communities, Nodes, and relevant projects guided by use cases.

Providers dashboard | **Use cases** | **Accounting reference framework**

useGalaxy.be
MGnify
RO-Crate

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